THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — MATHS · HARD LEVEL CHAPTER TEST SET 2

Chapter 9: **Symmetry** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. After a rotation of 200° a regular 9-gon coincides with itself. This rotation is 5 times the smallest rotation. The smallest rotation (degrees) is: 1

(a) 50 **(b)** 40 **(c)** 30 **(d)** 70

2. The number of letters among O, X, D, M, T with BOTH a vertical and a horizontal line of symmetry is: 1

(a) 4 **(b)** 3 **(c)** 2 **(d)** 1

3. The sum of the angles of rotation of a square and a regular octagon (in degrees) is: 1

(a) 145 **(b)** 125 **(c)** 135 **(d)** 165

4. A figure has rotational symmetry of order 5 and one of its lines of symmetry. The total number of lines of symmetry it must have is: 1

(a) 6 **(b)** 5 **(c)** 4 **(d)** 7

5. The reflection of a point (x, y) = (4, 2) in the y-axis (vertical line) has x-coordinate: 1

(a) -3 **(b)** 4 **(c)** -4 **(d)** -5

6. A regular polygon is rotated again and again through its smallest angle 30°. The number of turns needed to complete a full turn is: 1

(a) 11 **(b)** 14 **(c)** 13 **(d)** 12

7. A regular 8-gon has side 5 cm. Its perimeter (cm) is: 1

(a) 35 **(b)** 40 **(c)** 48 **(d)** 45

8. The image of a point 8 units from a mirror line is how far from the original point (units)? 1

(a) 8 **(b)** 24 **(c)** 16 **(d)** 17

9. A regular polygon has an angle of rotation 60° and its perimeter is 42 cm. The side length is (cm): 1

(a) 6 **(b)** 9 **(c)** 7 **(d)** 8

10. After a rotation of 210° a regular 12-gon coincides with itself. This rotation is 7 times the smallest rotation. The smallest rotation (degrees) is: 1

(a) 30 **(b)** 40 **(c)** 20 **(d)** 60

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. A figure rotates and fits after 45° and 135° turns but not at smaller turns. Find the order of symmetry from the smallest angle 45. 2

MASTERMIND

12. Reflect the point (4, 3) in 2
(a) the y-axis,
(b) the x-axis and give the coordinates.

13. A regular polygon has rotational symmetry with the smallest angle of 24° and each side 6 cm. Find the number of sides and its perimeter. 2

14. A figure rotates and fits after 60° and 240° turns but not at smaller turns. Find the order of symmetry from the smallest angle 60. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A designer wants a symmetrical logo with 8 identical petals. Find the angle between neighbouring petals, the number of lines of symmetry and the total of angles around the centre if 8 petals are placed. 3

16. Triangle ABC has A(3, 3), B(6, 3), C(4, 7) and is reflected in the vertical line x = 7. Find the images of A and B (the reflection of x is $2m - x$). 3

17. A regular 9-gon is rotated about its centre by 200° anticlockwise and the figure looks unchanged. Verify this using the order of symmetry and find the number of rotations of the smallest angle that make 200°. 3

OR

A designer wants a symmetrical logo with 9 identical petals. Find the angle between neighbouring petals, the number of lines of symmetry and the total of angles around the centre if 9 petals are placed. 3

18. Triangle ABC has A(1, 4), B(4, 4), C(2, 8) and is reflected in the vertical line x = 4. Find the images of A and B (the reflection of x is $2m - x$). 3

OR

A regular 10-gon is rotated about its centre by 144° anticlockwise and the figure looks unchanged. Verify this using the order of symmetry and find the number of rotations of the smallest angle that make 144°. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. For a regular 9-gon of side 3 cm: 5
(a) find the number of lines of symmetry,
(b) the angle of rotation,
(c) the perimeter,
(d) the angle turned by a vertex when the polygon is rotated 6 times through the smallest angle.

OR

Two regular polygons have 5 and 6 sides. Find their smallest angles of rotation, the difference, and the least angle (in degrees) that is a turn for both (the LCM of the two angles). 5

20. For a regular 10-gon of side 6 cm: 5
(a) find the number of lines of symmetry,
(b) the angle of rotation,
(c) the perimeter,
(d) the angle turned by a vertex when the polygon is rotated 3 times through the smallest angle.

OR

Two regular polygons have 6 and 9 sides. Find their smallest angles of rotation, the difference, and the least angle (in degrees) that is a turn for both (the LCM of the two angles). 5

END OF QUESTION PAPER — Total: 40 Marks